

Multiple Choice (10 marks)

1. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 3
 - (b) 4
 - (c) 7
 - (d) 15
2. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
3. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set
4. Statement 1| L_2 regularization of linear models tends to make models more sparse than L_1 regularization. Statement 2| Residual connections can be found in ResNets and Transformers.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Statement 1| The softmax function is commonly used in multiclass logistic regression. Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The computational complexity of gradient descent is polynomial in the number of features, D .
7. True or False: Word 2 Vec parameters are not initialized using a Restricted Boltzmann Machine.
8. True or False: Averaging the output of multiple decision trees helps decrease variance in the model's predictions.
9. True or False: The softmax function is commonly employed in multiclass logistic regression to convert logits into probabilities.
10. True or False: The BLEU metric primarily relies on precision, while the ROUGE metric focuses on recall in evaluating generated text.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the longest chain which can be formed from the given set of pairs.
12. Write a function to find maximum possible sum of disjoint pairs for the given array of integers and a number k .

End of Paper